# GROUPOIDS AND GROUPS OVER FIELDS ARE GEOMETRICALLY UNIBRANCH

THE STACKS PROJECT AUTHORS

*Editorial note.* No omission markers in-span. Dependence on the uncarded in-chapter move-point lemma is declared as external.

*Allowed standard prerequisites (editorial).*

- Descent of unibranchedness. Descent on Spaces, lemma descend-unibranch (cross-chapter, named). Reduces to the algebraically closed case along the covering.
- Restriction to a field extension. lemma-restrict-groupoid-on-field (same chapter, carded L26642). The covering properties used in the reduction.
- Minimal primes of a tensor product of strictly henselian local rings. More on Algebra, lemma minimal-primes-tensor-strictly-henselian (cross-chapter, named). Pairs of minimal primes of C correspond to pairs (p, p') in A.

*Source reference.* Original equation 01GQ: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 284–284. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04LS: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1015–1015. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04LM: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 880–880. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04LQ: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 937–937. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06DU: focused-seeds/geometry/S9-smg-groupoid-on-field-geometrically-irreducible/ref-tags.txt, source lines 5–5. Navigation pointer

*Date*: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06DX: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 853–853. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06DY: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 859–859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06DQ: focused-seeds/geometry/S9-smg-groupoid-on-field-geometrically-irreducible/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *In Situation 06DY the algebraic space $R$ is geometrically unibranch. In Situation 06DX the algebraic space $G$ is geometrically unibranch.*

*Proof.* Let $r \in |R|$. We have to show that $R$ is geometrically unibranch at $r$. Combining Lemma 04LQ with Descent on Spaces, Lemma 06DQ we see that it suffices to prove this in case $k$ is algebraically closed and $r$ comes from a morphism $r : \mathrm{Spec}(k) \to R$ such that $s \circ r$ and $t \circ r$ are automorphisms of $\mathrm{Spec}(k)$. By Lemma 04LS we reduce to the case that $r = e$ is the identity of $R$ and $k$ is algebraically closed.

Assume $r = e$ and $k$ is algebraically closed. Let $A = \mathcal{O}_{R,e}$ be the étale local ring of $R$ at $e$ and let $C = \mathcal{O}_{R \times_{s,U,t} R,(e,e)}$ be the étale local ring of $R \times_{s,U,t} R$ at $(e, e)$. By More on Algebra, Lemma 06DU the minimal prime ideals $\mathfrak{q}$ of $C$ correspond 1-to-1 to pairs of minimal primes $\mathfrak{p}, \mathfrak{p}' \subset A$. On the other hand, the composition law induces a flat ring map

$$\begin{array}{ccc} A & \xrightarrow{c^\sharp} & C \\ & & \uparrow \\ & & A \otimes_{s^\sharp,k,t^\sharp} A \end{array} \qquad \begin{array}{c} \mathfrak{q} \\ \vdash \\ \mathfrak{p} \otimes A + A \otimes \mathfrak{p}' \end{array}$$

Note that $(c^\sharp)^{-1}(\mathfrak{q})$ contains both $\mathfrak{p}$ and $\mathfrak{p}'$ as the diagrams

$$\begin{array}{ccc} A & \xrightarrow{c^\sharp} & C \\ \uparrow & & \uparrow \\ A \otimes_{s^\sharp,k} k & \xleftarrow{1 \otimes e^\sharp} & A \otimes_{s^\sharp,k,t^\sharp} A \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{c^\sharp} & C \\ \uparrow & & \uparrow \\ k \otimes_{k,t^\sharp} A & \xleftarrow{e^\sharp \otimes 1} & A \otimes_{s^\sharp,k,t^\sharp} A \end{array}$$

commute by (01GQ). Since $c^\sharp$ is flat (as $c$ is a flat morphism by Lemma 04LM), we see that $(c^\sharp)^{-1}(\mathfrak{q})$ is a minimal prime of $A$. Hence $\mathfrak{p} = (c^\sharp)^{-1}(\mathfrak{q}) = \mathfrak{p}'$. □